



PART - A

1. The difference between the compound interest and the simple interest earned at the end of 3rd year on a sum of money at a rate of 10% per annum is Rs. 77.5. What is the sum ?
 A) Rs. 3,500 B) Rs. 2,500
 C) Rs. 3,000 D) Rs. 2,000
2. Aamir and Birju can cut 5000 g of wood in 20 min. Birju and Charles can cut 5000 g of wood in 40 min. Charles and Aamir cut 5 kg of wood in 30 min. How much time Charles will take to cut 5 kg wood alone ?
 A) 120 min. B) 48 min.
 C) 240 min. D) 120/7 min.
3. An alloy contains copper and zinc in the ratio 5 : 3 and another contains copper and tin in the ratio 8 : 5. If equal weights of the two are melted together to form a 3rd alloy, find the weight of tin per kg in the new alloy.
 A) 40/129 B) 5/13
 C) 5/26 D) 28/5
4. x is a whole number. If the only common factors of x and x^2 are 1 and x , then x is _____.
 A) 1
 B) a perfect square
 C) an odd number
 D) a prime number
5. Line AB is 24 metres in length and is tangent to the inner one of the two concentric circles at point C. Points A and B lie on the circumference of the outer circle. It is known that the radii of the two circles are integers. The radius of the outer circle is
 A) 13 m B) 5 m
 C) 7 m D) 4 m

6. Monisha is working with a real estate agent to find a location for the kids' toy store she plans to open in her town. She is looking for a place that is either in the centre or not too far from the centre of town. It should also be attractive for the right kind of footfall too. Which of the following locations should Monisha's agent call to her attention ?
 A) a storefront in a new high-rise building near the train station in the center of town whose occupants are mainly young, childless professionals who use the train to commute to their offices each day
 B) a little shop three blocks away from the town's main street, located across the street from an elementary school and next door to an ice cream store
 C) a stand-alone storefront on a quiet residential street ten blocks away from the town's center
 D) a storefront in a small strip mall located on the outskirts of town that is also occupied by a pharmacy and a dry cleaner
7. Reading is a psycholinguistic guessing game. To read critically is a skill as it is a demanding process. One must slow down one's reading and, with a pencil in hand, perform specific operations on the text. Mark up the text with reactions, conclusions and questions. When one reads, one becomes an active participant.
 This paragraph best supports the statement that
 A) critical reading is a slow, dull, but essential process.
 B) the best critical reading happens at critical times in a person's life.
 C) readers should get in the habit of questioning the truth of what they read.
 D) critical reading requires thoughtful and careful attention.

SPACE FOR ROUGH WORK



8. Find the missing term :

60, 40, 55, 45, 50, 50, ?

- A) 45 B) 50
C) 55 D) 60

9. Find the missing alphabet :

T, r, O, m, J, ?

- A) h B) i
C) l D) g

10. Here are some words translated from an artificial language.

qmelaqali means fruitcake

qalitiimmo means cakewalk

useguamao means buttercup

Which word could mean "cupcake" ?

- A) qalitiqali B) amaotiimmo
C) pakitreft D) amaoqali

11. A man walks 5 km toward south and then turns to the right. After walking 3 km he turns to the left and walks 5 km. Now in which direction is he from the starting place ?

- A) West B) South
C) North-East D) South-West

12. If the consonants in the word 'DROVE' are first arranged alphabetically and the vowels are put in between two pairs of consonants in the alphabetical order, which of the following will be the fourth from the right end after the rearrangement ?

- A) D B) E
C) R D) O

13. There is a queue in a ticketing office. Amanda is 10th from the front while Murthy is 25th from behind and Marta is just in the middle of the two. If there be 50 persons in the queue. What position does Marta occupy from the front ?

- A) 16 B) 18
C) 15 D) 17

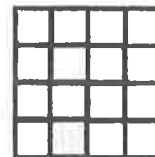
14. There are five janitors. Pali, Qureshi, Rohan, Sant and Timber. They all have a different height, Qureshi is shorter than only Timber and Sant is shorter than Pali and Qureshi. Who among them is the shortest ?

- A) Rohan
B) Sant
C) Pali
D) Data inadequate

15. A \$ B means A is the father of B; A # B means A is the sister of B; A * B means A is the daughter of B and A @ B means A is the brother of B. Which of the following indicates that M is the wife of Q ?

- A) Q \$ R # T @ M
B) Q \$ R @ T # M
C) Q \$ R * T # M
D) Q \$ R @ T * M

16. Count the number of squares in the given figure.



- A) 32 B) 30
C) 29 D) 28

SPACE FOR ROUGH WORK



17. Who was the Viceroy of India, when Quit India Resolution was passed in 1942 ?
A) Lord Linlithgow
B) Lord Wavell
C) Lord Willingdon
D) Lord Mountbatten
18. When was the East India Association set up ?
A) 1866 B) 1857
C) 1836 D) 1885
19. Who was the Spanish navigator who set out to discover India, but instead landed on the soil of America ?
A) Christopher Columbus
B) Vasco Da Gama
C) James Cook
D) None of above
20. Which dynasty was ruling over north India when Alexander the great invaded India ?
A) Gupta Dynasty
B) Maurya Dynasty
C) Sakya Dynasty
D) Nanda Dynasty
21. The roads of cities in the Indus Valley Civilization generally divided the city into
A) Rectangular Blocks
B) Circular Blocks
C) Triangular Blocks
D) None of above
22. In which year was Pulitzer Prize established ?
A) 1917 B) 1918
C) 1922 D) 1928
23. 'Kanchipuram' is in which of the following States ?
A) Andhra Pradesh
B) Orissa
C) Kerala
D) Tamil Nadu
24. Which of the following is not a chief organ of the United Nations Organisations ?
A) International Labour Organisation
B) Security Council
C) International Court of Justice
D) General Assembly
25. Which of the following is not a member of G-15 ?
A) Indonesia B) Malaysia
C) Columbia D) India
26. The group of metals Co, Ni, Fe may best called as
A) Transition metals
B) Main group metals
C) Alkali metals
D) Rare metals
27. Non stick cooking utensils are coated with
A) Black paint B) PVC
C) Teflon D) Polystyrene
28. The international township built near Pondicherry in India in coloration with UNESCO is called
A) Elbaville B) Auroville
C) Gayaville D) Broadway
29. Irvin sold a book at a profit of 12%. If Irvin had sold it for Rs. 18 more, then 18% would have been gained. Find the cost price.
A) Rs. 600 B) Rs. 300
C) Rs. 400 D) Rs. 200

SPACE FOR ROUGH WORK



30. In a group of 7 people, the average age is found to be 17 years. Two more people joined with an average age 19 years. One person left the group whose age was 25 years. What is the new average age of the group ?

A) 17.5 years B) 16.5 years
C) 18 years D) 16 years

31. A 300 metre long metro train crosses a platform in a metro station in 39 seconds while it crosses a lamp post in 18 seconds. What is the length of the platform ?

A) 250 metre B) 350 metre
C) 520 metre D) 300 metre

32. Assume that a sum of money is divided equally among n girls. Each girl will receive \$ 60. If another girl is added to the group and the sum is divided equally among all the girls, each child girl a \$ 50 share. What is the sum of money ?

A) \$ 3000 B) \$ 300
C) \$ 110 D) \$ 10

33. A tank can be filled by one tap in 10 minutes and by another in 30 minutes. Both the taps are kept open for 5 minutes and then the first one is shut off. In how many minutes more is the tank completely filled ?

A) 5 B) 7.5
C) 10 D) 12

Direction for Questions 34 – 38 : From the four choices provided, choose the analogy that is most similar to the one in the question.

34. Wealth : Poverty

A) part : whole B) good : excellent
C) prodigal : chary D) wicked : sinful

35. Misfortune : Catastrophe

A) miniature : big
B) limited : infinite
C) knowledge : learning
D) generosity : parsimony

36. Molecule : Atoms

A) family : sisters B) light : bulb
C) tissue : cells D) body : limb

37. Limp : Walk

A) flap : fly B) run : race
C) stutter : talk D) chew : digest

38. Riddle : Solve

A) mirage : illusion B) joke : amuse
C) tangle : unravel D) target : aim

39. Fact 1 : All chickens are birds.

Fact 2 : Some chickens are hens.

Fact 3 : Female birds lay eggs.

If the first three statements are facts, which of the following statements must also be a fact ?

I. All birds lay eggs.
II. Hens are birds.
III. Some chickens are not hens.

A) II only
B) II and III only
C) I, II and III
D) None of the statements is a known fact

40. What is the most essential thing for election ?

A) President B) Voter
C) November D) Nation

SPACE FOR ROUGH WORK



41. What is the most essential thing for ovation ?

- A) outburst B) bravo
C) applause D) encore

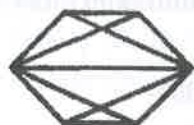
42. Introducing a man to her husband, a woman said, "His brother's father is the only son of my grandfather." How is the woman related to this man ?

- A) Mother B) Aunt
C) Sister D) Daughter

43. Pointing to a photograph, a man said, "I have no brother or sister but that man's father is my father's son." Whose photograph was it ?

- A) His Own
B) His Son
C) His Father
D) His Grandfather

44. Find the number of quadrilaterals in the given figure.



- A) 6 B) 7
C) 9 D) 11

45. Count the number of convex pentagons in the adjoining figure.



- A) 16 B) 12
C) 8 D) 4

Read the information given below and then answer questions 46 – 49 : There is a family of six people whose nick names are Pat, Qat, Rat, Sat, Tat and Uat. Their professions are Engineer, Doctor, Teacher, Salesman, Manager and Lawyer.

There are two married couples in the family.

The Manager is the grandfather of Uat, who is an Engineer

Rat, the Salesman, is married to the lady Teacher
Qat is the mother of Uat and Tat.

The Doctor, Sat is married to the Manager.

46. How many male members are there in the family ?

- A) Two B) Three
C) Four D) Data Inadequate

47. What is the profession of Pat ?

- A) Lawyer
B) Lawyer or Teacher
C) Manager
D) None of these

48. Who are the two married couples in the family ?

- A) Pat-Qat and Sat-Rat
B) Rat-Uat and Sat-Tat
C) Pat-Tat and Sat-Rat
D) Pat-Sat and Rat-Qat

49. How Pat is related to Tat ?

- A) Father B) Grandfather
C) Mother D) Grandmother

50. Shirin went to a bakery and bought items worth Rs. 25, out of which 30 paise went on sales tax on taxable purchases. If the tax rate was 6%, then what was the cost of the tax free items ?

- A) Rs. 12 B) Rs. 19.70
C) Rs. 19.10 D) Rs. 18.80

SPACE FOR ROUGH WORK



51. The rate of simple interest on a sum of money is 6 per cent per annum for the first 3 years, 8 per cent per annum for the next 5 years and 10 per cent per annum for the period beyond 8 years. If the simple interest accrued by the sum for a total period of 10 years is Rs. 1,560, what is the sum ?
- A) Rs. 1,500
B) Rs. 3,000
C) Rs. 2,000
D) Data Inadequate
52. Adrian starts a start-up with a capital of Rs. 85,000. Brian joins in the start-up with Rs. 42,500 aftersometime. For how much period does Brian join, if the profits at the end of the year are divided in the ratio of 3 : 1 ?
- A) 5 months B) 6 months
C) 7 months D) 8 months
53. A car travels at an average of 50 miles per hour for 2.5 hours and then travels at a speed of 70 miles per hour for 1.5 hours. How far did the car travel in the entire 4 hours ?
- A) 210 miles B) 230 miles
C) 250 miles D) 260 miles
54. By selling 45 limes for Rs. 40, a woman loses 20%. How many should she sell for Rs. 24 to gain 20 % in the transaction ?
- A) 16 B) 18
C) 20 D) 22
55. The prestigious Ramon Magsaysay Award was conferred upon Ms. Kiran Bedi for her excellent contribution to which of the following fields ?
- A) Literature
B) Community Welfare
C) Government Service
D) Journalism
56. Who among the following is not a recipient of 'Dada Saheb Phalke' Award ?
- A) Ramanand Sagar
B) Raj Kapoor
C) V. Shantaram
D) Ashok Kumar
57. What is part of a database that holds only one type of information ?
- A) Report B) Field
C) Record D) File
58. '.JPG' extension refers usually to what kind of file ?
- A) System file
B) Animation/movie file
C) MS Encarta document
D) Image file
59. Which of the following is not written by Munshi Premchand ?
- A) Gaban B) Godan
C) Guide D) Manasorovar
60. The famous Dilwara Temples are situated in
- A) Uttar Pradesh B) Rajasthan
C) Maharashtra D) Madhya Pradesh

SPACE FOR ROUGH WORK



PART - B

61. What will be the output of following ?

```
main()
{
    Static int a = 3;
    Printf("%d",a- -);
    If (a)
        main();
}
```

- A) 3
- B) 3 2 1
- C) 3 3 3
- D) Program will fall in continuous loop and print 3

62. The height of a binary tree is the maximum number of edges in any root to leaf path. The maximum number of nodes in a binary tree of height h is

- A) 2^h
- B) $2^{h-1} - 1$
- C) $2^{h+1} - 1$
- D) 2^{h+1}

63. Web links are stored within the page itself and when you wish to 'jump' to the page that is linked, we select the hotspot or anchor. This technique is called

- A) Hypertext
- B) Hypermedia
- C) Both A) and B)
- D) Anchoring

64. If the objects focus on the problem domain, then we are concerned with

- A) Object Oriented Analysis
- B) Object Oriented Design
- C) Object Oriented Analysis and Design
- D) None of the above

65. Which of the following is not a form of main memory ?

- A) Instruction cache
- B) Instruction register
- C) Instruction opcode
- D) Translation look-aside buffer

66. Which of the following is useful in traversing a given graph by breadth first search ?

- A) Stack
- B) Set
- C) List
- D) Queue

67. M is a square matrix of order ' n ' and its determinant value is 5. If all the elements of M are multiplied by 2, its determinant value becomes 40. The value of ' n ' is

- A) 2
- B) 3
- C) 5
- D) 4

68. Is null an object ?

- A) Yes
- B) No
- C) Sometimes yes
- D) None of these

69. Which of the following scheduling algorithms could result in starvation ?

- A) Priority
- B) Round Robin
- C) FCFS
- D) None of the above

70. Which of the following is a desirable property of module ?

- A) Independency
- B) Low cohesiveness
- C) High coupling
- D) Multifunctional

SPACE FOR ROUGH WORK

71. Which of these events will be generated if we close an applet's window ?
 A) ActionEvent
 B) ComponentEvent
 C) AdjustmentEvent
 D) WindowEvent
72. If queue is implemented using arrays, what would be the worst run time complexity of queue and dequeue operations ?
 A) $O(n)$, $O(n)$
 B) $O(n)$, $O(1)$
 C) $O(1)$, $O(n)$
 D) $O(1)$, $O(1)$
73. The following program fragment prints
 inti = 5;
 do { putchar(i+100); printf("%d", i--); }
 while (i);
 A) i5h4g3f2e1
 B) 14h3g2f1e0
 C) an error message
 D) none of the above
74. The running time of an algorithm $T(n)$, where 'n' is the input size, is given by
 $T(n) = 8T(n/2) + qn$, if $n > 1$
 $= p$, if $n = 1$
 Where p, q are constants. The order of this algorithm is
 A) n^2
 B) n^n
 C) n^3
 D) n
75. Consider a system having 'm' resources of the same type. These resources are shared by 3 processes A, B, C; which have peak time demands of 3, 4, 6 respectively. The minimum value of 'm' that ensures that deadlock will never occur is
 A) 11
 B) 12
 C) 13
 D) 14
76. For a database relation $R(a, b, c, d)$ where the domains of a, b, c and d include only atomic values, only the following functional dependencies and those that can be inferred from them hold.
 $a \rightarrow c$
 $b \rightarrow d$
 The relation is in
 A) First normal form but not in second normal form
 B) Second normal form but not in third normal form
 C) Third normal form
 D) None of the above
77. The convergence of the bisection method is
 A) Cubic
 B) Quadratic
 C) Linear
 D) None
78. A pipeline is having speed up factor as 10 and operating with efficiency of 80%. What will be the number of stages in the pipeline?
 A) 10
 B) 8
 C) 13
 D) None

SPACE FOR ROUGH WORK



79. In a ripple counter using edge-triggered JK flip-flops, the pulse input is applied to

- A) Clock input of all flip-flops
- B) J and K input of one flip-flop
- C) J and K input of all flip-flops
- D) Clock input of one flip-flop

80. The number of possible binary trees with 4 nodes is

- A) 12
- B) 13
- C) 14
- D) 15

81. A program P calls two subprograms P1 and P2. P1 can fail 50% time and P2 can fail 40% times. The program P can fail

- A) 50%
- B) 10%
- C) 60%
- D) 70%

82. In time division switches if each memory access takes 100 ns and one frame period is 125 μ s, then the maximum number of lines that can be supported is

- A) 625 lines
- B) 1250 lines
- C) 2300 lines
- D) 318 lines

83. Which two are valid constructors for Thread ?

1. Thread(Runnable r, String name)
2. Thread()
3. Thread(int priority)
4. Thread(Runnable r, ThreadGroup g)
5. Thread(Runnable r, int priority)

- A) 1 and 3
- B) 2 and 4
- C) 1 and 2
- D) 2 and 5

84. Which of the following object types are generally autonomous, meaning that they can exhibit some behavior without being operated upon by another object.

- A) Passive
- B) Active
- C) Both A) and B)
- D) None of the mentioned

85. The address of a class B Host is to be split into subnets with a 3 - bit subnet number. What is the maximum number of subnets and maximum number of hosts in each subnet ?

- A) 8 subnets and 262142 hosts
- B) 6 subnets and 262142 hosts
- C) 6 subnets and 1022 hosts
- D) 8 subnets and 1024 hosts

86. Which level of RAID refers to disk mirroring with block striping ?

- A) RAID level 1
- B) RAID level 2
- C) RAID level 0
- D) RAID level 3

87. Which of the following desired features are beyond the capability of relational algebra ?

- A) Aggregate Computation
- B) Multiplication
- C) Finding transitive closure
- D) All of the above

88. What is 'Basis of Encapsulation' ?

- A) object
- B) class
- C) method
- D) all of the mentioned

SPACE FOR ROUGH WORK



89. Consider the following declaration.

int a, *b = &a, **c = &b;

a = 4;

**c = 5;

If the statement

b = (int *)**c

Is appended to the above program fragment then

- A) Value of b becomes 5
- B) Value of b will be the address of c
- C) Value of b is unaffected
- D) None of these

90. How many 2-input multiplexers are required to construct a 2^{10} input multiplexer?

- A) 1023
- B) 31
- C) 10
- D) 127

91. When transaction T_i requests a data item currently held by T_j , T_i is allowed to wait only if it has a time stamp smaller than that of T_j (that is, T_i is older than T_j). Otherwise, T_i is rolled back (dies). This is

- A) Wait-die
- B) Wait-wound
- C) Wound-wait
- D) Wait

92. The maximum data rate of a channel of 3000-Hz bandwidth and SNR of 30 db is

- A) 60000
- B) 15000
- C) 30000
- D) 3000

93. The number of columns in a stateable for a sequential circuit with 'm' flip-flops and 'n' input is

- A) $m + n$
- B) $m + 2n$
- C) $2m + n$
- D) $2m + 2n$

94. The solution of the recurrence relation $a_r = a_{r-1} + 2a_{r-2}$ with $a_0 = 2$, $a_1 = 7$ is

- A) $a_r = (3)^r + (1)^r$
- B) $2a_r = (2)^r/3 - (1)^r$
- C) $a_r = 3^{r+1} - (-1)^r$
- D) $a_r = 3(2)^r - (-1)^r$

95. In a 10-bit computer instruction format, the size of address field is 3-bits. The computer uses expanding OP code technique and has 4 two-address instructions and 16 one-address instructions. The number of zero address instructions it can support is

- A) 256
- B) 356
- C) 640
- D) 756

96. The average search time for hashing with linear probing will be less if the load factor

- A) Is far less than one
- B) Equals one
- C) Is far greater than one
- D) None of the above

SPACE FOR ROUGH WORK



PART - C

97. The question is based on the following program fragment.

```
f (int Y[10], int x){
    int u, j, k;
    i = 0; j = 9;
    do {
        k = (i+j) / 2;
        if (Y[k] < x) i = k; else j = k;
    } while ( (Y[k] != x) && (i < j) );
    if (Y[k] == x) printf ("x is in the array.");
    else printf ("x is not in the array.");
}
```

On which of the following contents of 'Y' and 'x' does the program fail ?

- A) Y is [1 2 3 4 5 6 7 8 9 10] and $x < 10$
- B) Y is [1 3 5 7 9 11 13 15 17 19] and $x < 1$
- C) Y is [2 2 2 2 2 2 2 2 2 2] and $x > 2$
- D) Y is [2 4 6 8 10 12 14 16 18 20] and $2 < x < 20$ and 'x' is even

98. An algorithm is made up of two modules M1 and M2. If order of M1 is $f(n)$ and M2 is $g(n)$ then the order of algorithm is

- A) $\max(f(n), g(n))$
- B) $\min(f(n), g(n))$
- C) $f(n) + g(n)$
- D) $f(n) \times g(n)$

99. By open domain CASE tools we mean

- A) tools available in open domain
- B) software packages which can be downloaded from the internet
- C) software packages to aid each phase of the systems analysis and design which can be downloaded free of cost from the internet
- D) source codes of CASE tools

100. Bit stuffing refers to

- A) Inserting a '0' in user data stream to differentiate it with a flag
- B) Inserting a '0' in flag stream to avoid ambiguity
- C) Appending a nibble to the flag sequence
- D) Appending a nibble to the user data stream

101. Assume transaction A holds a shared lock R. If transaction B also requests for a shared lock on R. It will

- A) result in deadlock situation
- B) immediately be granted
- C) immediately be rejected
- D) be granted as soon as it is released by A

102. Given relations $R(w, x)$ and $S(y, z)$, the result of $\text{SELECT DISTINCT } w, x \text{ from } R, S$

- A) R has no duplicates and S is non-empty
- B) R and S have no duplicates
- C) S has no duplicates and R is non-empty
- D) R and S has the same number of tuples

SPACE FOR ROUGH WORK

103. A decimal has 25 digits. The number of bits needed for its equivalent binary representation is approximately

- A) 50
- B) 74
- C) 40
- D) 60

104. Match List-I with List-II and select the correct answer using the codes given below the Lists :

List - I

a. 0-address instruction

b. 1-address instruction

c. 2-address instruction

d. 3-address instruction

List - II

1. $T = TOP(T - 1)$

2. $Y = Y + X$

3. $Y = A - B$

4. $ACC = ACC - X$

Codes :

- | | a | b | c | d |
|----|---|---|---|---|
| A) | 1 | 2 | 3 | 4 |
| B) | 3 | 2 | 4 | 1 |
| C) | 2 | 3 | 1 | 4 |
| D) | 1 | 4 | 2 | 3 |

105. Which of the following definitions generates the same languages as L, where $L = \{x^n y^n, n \geq 1\}$

- i. $E \rightarrow xEy \mid xy$
- ii. $xy \mid x^+xyy^+$
- iii. x^+y^+

- A) (i) only
- B) (i) and (ii) only
- C) (ii) and (iii) only
- D) (ii) only

106. The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 records per page, with 1 free main memory frame is recorded as follows. What is the number of Page Faults?

0100, 0200, 0430, 0499, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0370.

- A) 15, 4
- B) 6, 4
- C) 7, 2
- D) 4, 6

107. If the channel is band limited to 6 kHz and signal to noise ratio is 16, what would be the capacity of channel ?

- A) 16.15 kbps
- B) 23.24 kbps
- C) 40.12 kbps
- D) 24.74 kbps

108. At 100% modulation, the power in each sideband is _____ of that of carrier.

- A) 50%
- B) 40%
- C) 60%
- D) 25%

SPACE FOR ROUGH WORK



PART - D

109. Twelve $1\ \Omega$ resistances are used as edges to form a cube. The resistance between two diagonally opposite corners of the cube is
- A) $\frac{5}{6}\ \Omega$
- B) $\frac{1}{6}\ \Omega$
- C) $\frac{6}{5}\ \Omega$
- D) $\frac{3}{2}\ \Omega$
110. An ideal op-amp is an ideal
- A) voltage controlled current source
- B) voltage controlled voltage source
- C) current controlled current source
- D) current controlled voltage source
111. The final value theorem is used to find the
- A) Steady state value of the system output
- B) Initial value of the system output
- C) Transient behavior of the system output
- D) None of these

112. For the discrete signal $x[n] = a^n u[n]$, $a > 0$ the z-transform is
- A) $\frac{(z+a)}{z}$
- B) $\frac{(z-a)}{z}$
- C) $\frac{z}{(z-a)}$
- D) $\frac{z}{(z+a)}$
113. For a periodic signal $v(t) = 30 \sin 100t + 10 \cos 300t + 6 \sin (500t + \pi/4)$, the fundamental frequency in rad/s
- A) 100
- B) 300
- C) 500
- D) None of these
114. A solution for the differential equation $x'(t) + 2x(t) = \delta(t)$ with initial condition $x(0) = 0$
- A) $e^{-2t} u(t)$
- B) $e^{2t} u(t)$
- C) $e^{-t} u(t)$
- D) $e^t u(t)$
115. If the number of bits per sample in a PCM system is increased from a n to $n+1$, the improvement in signal to quantization noise ratio will be
- A) 3 dB
- B) 6 dB
- C) $2n$ dB
- D) n dB

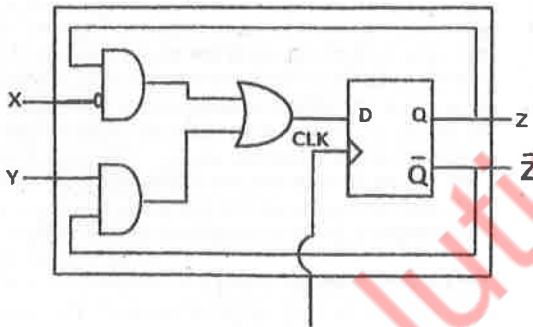
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116. A carrier $A_c \cos(\omega_c t)$ is frequency modulated by a signal $E_m \cos(\omega_m t)$. The modulation index is m_f . The expression for the resulting FM signal is

- A) $A_c \cos[\omega_c t + m_f \sin(\omega_m t)]$
- B) $A_c \cos[\omega_c t + m_f \cos(\omega_m t)]$
- C) $A_c \cos[\omega_c t + \pi m_f \sin \omega_m t]$
- D) $A_c \cos[\omega_c t + 2\pi m_f E_m \cos(\omega_m t) / \omega_m]$

117. A sequential circuit using D flip-flop and logic gates is shown in Figure, where X and Y are the inputs and Z is the output. The circuit is



- A) S - R Flip-Flop with inputs $X = R$ and $Y = S$
- B) S - R Flip-Flop with inputs $X = S$ and $Y = R$
- C) J - K Flip-Flop with inputs $X = J$ and $Y = K$
- D) J - K Flip-Flop with inputs $X = K$ and $Y = J$

118. A 4 bit ripple counter and a 4 bit synchronous counter are made using flip-flops having a propagation delay of 10 ns each. If the worst case delay in the ripple counter and the synchronous counter be R and S respectively, then

- A) $R = 10 \text{ ns}, S = 40 \text{ ns}$
- B) $R = 40 \text{ ns}, S = 10 \text{ ns}$
- C) $R = 10 \text{ ns}, S = 30 \text{ ns}$
- D) $R = 30 \text{ ns}, S = 10 \text{ ns}$

119. In MOSFET fabrication, the channel length is defined during the process of

- A) Isolation oxide growth
- B) Channel stop implantation
- C) Poly-silicon gate patterning
- D) Lithography step leading to the contact pad

120. The open-loop transfer function of a feedback control system is $G(s).H(s) = 1/(s + 1)^3$. The gain margin of the system is

- A) 2
- B) 4
- C) 8
- D) 16

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